

Reviewer 2: threshold sensitivity and the vaccine-antigen benchmark

The combined R2.1 / R2.3 analysis has been run; §01 answers each request in turn; §05 carries the scientific review. Four of the draft plan's ten sensitivity scenarios do not work — three change nothing and one empties the catalogue — and the antigen benchmark returns “excluded” in every cell. Replacements are proposed and tested. Six wording corrections are copy-ready, five already carried in this report.

From Andrew Balmer

Analysis analysis/ on r2-sensitivity-analysis

App pf-peptidefilter.streamlit.app

Verified against live prod 5861f79, 4 Sep 2026

THE ANSWERS, IN SIX LINES

R2.1

None of the eight benchmark antigens is retained, under any of the 15 configurations. Strain conservation is the criterion responsible in every case.

R2.3

The catalogue ranges **176 to 1,142 genes** around the published 253. Most sensitive to expression and conservation; least to indel frequency.

Why

Five of the eight antigens sit in the bottom 1% of the genome for conservation, and a sixth is just outside at 2.03%. Not a life-stage artefact — a sporozoite-stage filter still retains none of them, and removing expression and orthology altogether still retains none.

Plan changes

Four of the plan's ten scenarios measure nothing or empty the catalogue. Replacements are tested and selectable in the app. Four departures in total, each with its evidence in §01.

For the paper

Six wording corrections, **five of them already in print**. Copy-ready in MANUSCRIPT-CORRECTIONS.md; the reasoning is in §04.

Confidence

Every figure comes from the app's own `filter_datasets().validate()` reproduces the six published preprint numbers before any script runs, and aborts on drift.

How this is organised, and what to read

The plan comes first, then the two analyses in full. §04 **gathers every change the manuscript needs into one place**, so the text edits can be worked through in a single pass. §05 onward is additional.

Plain

What the analysis did, in ordinary language, with the one worked example that governs everything else. **Start here.**

§01

Every item in the plan, numbered, with where each one landed — then the four places the work departs from it, each with the run that forced it.

§02

R2.3 in full — the fifteen scenarios, what each one moves, and the ones that had to be replaced.

[§03](#)

R2.1 in full — the eight antigens, why each fails, and how far each sits from selectable.

[§04](#)

Every manuscript change in one place — the six wording corrections, the redrafted methods paragraph, and the Results passage drafted for Step 20. Sentence as published, replacement beneath.

[§05](#)

Additional comments, not part of the plan — how the numbers were produced and checked, then eight points that came out of the work.

[§06-07](#)

The outputs proposed for the supplement, and the decisions still open.

[Plain](#) What this analysis did, in plain terms

1 • What was asked. Reviewer 2 asked for two related checks. R2.3: is the catalogue Pf-PeptideFilter produces stable when the main thresholds move, or an artefact of one hand-picked setting? R2.1: how do well-known malaria vaccine antigens behave under the same filtering logic? Both are answered in one run of 15 configurations, as the plan specified.

2 • What was run. The reference catalogue: The stricter M253 setting is the app default plus liver-stage day-4 expression and *P. vivax* orthology. It retains 253 genes and 10,088 peptides. The app's default catalogue retains 1,142 genes and 41,332 peptides, and M253 is a strict subset of that broader default output.

The fifteen configurations. Fifteen configurations were run. Each changes one filter from the M253 reference, then records catalogue size and overlap with both M253 and the app default at gene and peptide level. Four of the ten draft scenarios were rejected for testing nothing useful — the two human-identity settings ($\leq 90\%$ and $\leq 70\%$) were inert, an indel-stringent setting was inert, and requiring perfect conservation emptied the catalogue — and a fifth was replaced because its value cannot be selected in the app. Empirically meaningful, on-grid alternatives took their place, with the rejected options kept in `rejected_scenarios.csv`.

The antigen benchmark. The eight requested antigens were resolved to stable PlasmoDB gene IDs and checked in every run: CSP, AMA1, RH5, TRAP, CeITOS,

EBA175, MSP1 and LSA3. “Excluded” means the gene did not survive all active filters for that run. The antigen table reports the slider setting being tested, the underlying fraction of peptides passing, and a genome-wide percentile rank, so a near miss can be told from a severe failure.

3 • What was found. Single-filter catalogue sizes range from 176 to 1,142 genes around the 253-gene M253 baseline. Relaxing filters adds genes, tightening removes them, and combined relaxations are super-additive because some genes fail more than one filter at once. None of the eight benchmark antigens re-enters the catalogue in any of the 15 scenarios. All eight fail the configured strain-conservation criterion, five of the eight sit in the bottom 1% of all genes on the conservation percentile, and a sixth is just outside at 2.03%. The tool rejects them because they fit this conservation-first logic poorly.

4 • What this cannot tell us. The benchmark does not test immune protection, epitope presentation, formulation, clinical performance, or whether a rejected antigen is a bad target. The life-stage and orthology filters are also not equally fair to all eight, because the panel mixes pre-erythrocytic and blood-stage biology and includes *P. falciparum*-restricted targets. The conservation conclusion is the robust one: even with expression and orthology relaxed, or swapped to a sporozoite-stage setting, the eight are still not retained.

01 Response to each plan item

Each item in the plan, numbered, with what was produced and where it lives. Quotations are verbatim from Steps 1–20. Four were answered *differently* from the way they were specified; the reasons follow the table.

Reviewer 2 asked for two things: **R2.1**, benchmark the licensed and leading malaria vaccine antigens, and **R2.3**, add a threshold sensitivity analysis. One set of 15 configurations carries both answers, as the plan specified.

#	ASKED FOR	DELIVERED, AND WHERE	STATUS	WHAT IT SHOWED
1	“one main supplementary sensitivity table covering the baseline and relaxed/stringent scenarios”	15 rows × 22 columns. supplementary_table_A_sensitivity.csv; trimmed to 8 columns in §02	ANSWERED	Catalogue ranges 176–1,142 genes across single-filter moves from a 253-gene baseline
2	“comparison with both the default catalogue and the 253 gene catalogue”	Shared count, % retained and Jaccard against both references, at gene <i>and</i> peptide level — 12 of the 22 columns · §02	ANSWERED	M253 is a strict subset of APP: 253 of 1,142 genes, Jaccard 0.222

#	ASKED FOR	DELIVERED, AND WHERE	STATUS	WHAT IT SHOWED
3	“antigen retention for every scenario (column)”	benchmark_antigens_retained, column 22 of Table A; also the full 8 × 15 grid in antigen_retention_matrix.csv · §03	ANSWERED	Reads none retained in all 15 rows, spelled that way because pandas reads None back as missing. Dropped from the display table for that reason; it is in the CSV
4	“one smaller antigen-focused supplementary table explaining when each antigen gene passes or fails”	8 rows × 32 columns. supplementary_table_B_antigens.csv; 9 columns shown in §03	ANSWERED	All eight fail conservation; M253_exclusion_reason lists every filter each one fails, not just the first
5	“three settings for each filter: relaxed, default and stringent, with a short rationale for each choice”	Five filters × three settings, rationale per value in §02	ANSWERED DIFFERENTLY	No stringent direction on any of the indel filter’s four controls , so it gets a relaxation-only pair — see §02
6	“vary one filter at a time, keeping all other filters fixed”	All 15 scenarios differ from M253 in one filter only · §02	ANSWERED	Holds for all 15, though three rows change two settings of the human filter — see the departures below
7	“two reference baselines” — application defaults, and “the 253 gene example, using the exact configuration that generated that result”	config_APP.json, config_M253.json, both replayable · §02	ANSWERED	APP 1,142 / 41,332; M253 253 / 10,088. “Default + activating the last two filters” is exactly right and reproduces
8	“For every run, we record the number of retained genes and peptides and the overlap with both reference outputs”	Columns 4–17 of Table A; plus per-run gene and peptide catalogues, 45 files in run_outputs/ · §02	ANSWERED	All 15 reproduce from their frozen configs; containment verified by set inclusion at both levels
9	“check if the antigens requested by the reviewer are retained or discarded: CSP, AMA1, RH5, TRAP, CeITOS, EBA175, MSP1 and LSA3”	All eight, identified by stable PlasmoDB IDs, checked in all 15 runs · §03	ANSWERED	None retained in any of the 15. Identifiers are not substring matches — MTRAP, TRAPPC2/4/5, MSP10 and MSP11 are separate rows
10	“the interpretation should be that we are testing how established vaccine antigens behave under the specific filtering logic of the app, not if they are good or bad vaccine targets” — flagged Important	§03 opening and closing paragraphs	ANSWERED	§03 states it explicitly and the “working as intended” reading is marked as interpretation rather than output

#	ASKED FOR	DELIVERED, AND WHERE	STATUS	WHAT IT SHOWED
11	“check the scenario definition and the rationale and more importantly if they make sense empirically (e.g., perhaps one configuration removes all genes which is obviously meaningless)”	rejected_scenarios.csv — five scenarios tested and discarded with reasons · §02	ANSWERED	Where the real findings came from. Four of the plan’s ten scenarios do nothing or empty the catalogue, including the case it anticipated (conservation = 1.00 → 0 genes)
12	Step 10 — the per-run output list (catalogue size; shared/%Jaccard vs M253 and vs APP at both levels; direction-of-change counts)	All 18 specified quantities, as 22 columns with scenario/filter/setting/antigens · §02	ANSWERED	Every numeric cell recomputes independently
13	Step 12 — “benchmark-antigen identifier table” as a named output	Folded into Table B: antigen, gene_id, gene_name_in_app, n_peptides · §03	ANSWERED	Not a separate file; the four identifier columns lead Table B instead
14	Step 14 — “report the relevant percentage of peptides passing rather than only a binary PASS/FAIL”, to distinguish “94% of peptides pass a 95% requirement” from “only 20% pass”	*_true_pct_peptides_passing for all three peptide-extrapolated filters, alongside *_gene_rule_pct_tolerated · §03	ANSWERED	All three peptide-extrapolated filters. Conservation, % of peptides passing: AMA1 27.4, TRAP 22.8, CelTOS 50.0, MSP1 54.7, LSA3 69.7, EBA175 72.0, CSP 74.4, RH5 80.8, against a 95% requirement
15	Step 14 — “final M253 status” and “final APP status” per antigen	M253_status, APP_status · §03	ANSWERED	“excluded” in all 16 cells
16	Steps 8, 9 and 15 — “The analysis therefore requires 12 unique runs ” / “should have 12 rows ”	15 runs · §01	ANSWERED DIFFERENTLY	12 + 3 identity-only rows. The count difference is entirely the human filter: see departure 1
17	Step 17 / Step 20 — an 8 × 12 retained/excluded matrix, “particularly well as a simple supplementary heatmap”, to show whether an antigen “re-enters the catalogue under a modest relaxation”	Replaced by the distance-to-threshold figure. Matrix still shipped as antigen_retention_matrix.csv (8 × 15) · §03	ANSWERED DIFFERENTLY	Every cell is “excluded”, so the heatmap is one flat block. The figure answers the same question: how far from selectable is each antigen

#	ASKED FOR	DELIVERED, AND WHERE	STATUS	WHAT IT SHOWED
18	Step 19 — reproducibility outputs: a frozen config per run, gene- and peptide-level output per run, the identifier table, both supplementary tables, the retention matrix, and the scripts	15 configs, 45 run files, both tables, the matrix, the analysis/verification/rendering scripts, provenance.txt, COLUMN-DEFINITIONS.md, and the source of this report · §06	ANSWERED	All 15 configs replay from disk to their table rows
19	Step 19 / Step 2 — “The scenario definitions should be frozen before the filtering outputs are inspected”	Scenarios were frozen, run, and then four were replaced after inspection · §01	ANSWERED DIFFERENTLY	Contradicted by the instruction to check the scenarios empirically — see departure 3
20	Step 20 — a short Results summary with five named components	Proposed passage at the end of this section · §04	ANSWERED	Drafted here for the first time, all five components covered, clearly marked as proposed text
21	Step 6 — “The 5-CPM threshold is a prespecified sensitivity-analysis value , not a biologically validated boundary ... It should be described as such”	Stated in §02 alongside the expression scenarios	ANSWERED	Stated explicitly wherever the 5 CPM value is used
22	Step 18 — recovery of an antigen “provides a useful face-validity benchmark ... but does not demonstrate that its retained peptides are immunogenic or protective”; an antigen “may fail because it is highly polymorphic, despite having demonstrated biological or immunological relevance”	§03	ANSWERED	Both sentences now carried explicitly; the second is the substance of the result
23	The invitation to “modify the plan or suggest changes”	Four departures, below, each with the evidence that forced it · §01	ANSWERED	Three of the four were forced by the empirical check the plan asked for

The four departures from the plan

1 — The human filter varies alignment length, not identity, and gains three identity-only rows.

Step 3 is explicit: “For the principal sensitivity analysis, only the **maximum percentage identity** will be varied. Alignment length, the joint rule and the gene-level peptide requirement will remain fixed at their M253 values”, and under “Important design choice”: “Do **not** simultaneously

change identity, alignment length and gene-peptide proportion. Doing so would make it impossible to determine which component of the human-similarity filter caused the observed change.”

The delivered analysis does both prohibited things, because the scenarios as specified measure nothing. With everything else held at M253, the identity threshold returns **exactly 253 genes at $\leq 60\%$, $\leq 70\%$, $\leq 80\%$, $\leq 90\%$ and $\leq 100\%$** : the ≤ 15 aa alignment-length criterion binds across that whole range, so identity never gets to act. Varying length instead gives 183 / 253 / 376 genes at ≤ 12 / ≤ 15 / ≤ 18 aa.

On why ± 3 aa, which the plan specified for identity but not for length: it is the smallest symmetric step that moves the catalogue materially in both directions (-28% and $+49\%$) while leaving the criterion recognisable. At ≤ 20 aa, the top of the slider, the filter excludes nothing; at ± 1 aa too few genes move to separate a real dependence from granularity in the alignment lengths. Like the plan’s own ± 10 points on identity, it is a defensible span rather than a validated boundary. Switching the length filter off, which the app allows, makes identity informative again: 506 / 558 / 587 genes at $\leq 70\%$ / $\leq 80\%$ / $\leq 90\%$. Separating the arms meets Step 3’s attribution concern by construction, since each row moves one component. The largest departure of the four.

2 — The indel filter has no stringent arm. All four of its controls were checked. The gene rule already sits at its 100% maximum. The frequency threshold is inert from 0.05 down to 0.00, because every gene at or below 0.05 sits at exactly 0.0 and the next distinct value in the data is 0.057. And restricting *which* indels are counted **relaxes** the filter, since the metric is a maximum over a smaller set: “Only Insertions” gives 280 genes, “Only Deletions” 260, “Only Frameshifts” 294, insertions+frameshifts 297, against 253 at baseline. Relaxation-only is a property of the filter at this configuration, and belongs in the Results as a finding.

3 — The scenarios were revised after seeing their output. Steps 2 and 19 ask for the definitions to be fixed before the outputs are inspected, but the plan also asks for the opposite: *“check the scenario definition and the rationale and more importantly if they make sense empirically (e.g., perhaps one configuration removes all genes which is obviously meaningless)”*. Checking requires looking. The scenarios were defined, run, and four replaced once their outputs were seen — three for being inert, one for emptying the catalogue. `rejected_scenarios.csv` records all five with reasons.

4 — The conservation-stringent arm moved from a threshold to a gene rule. The plan specified 1.00, which empties the catalogue: only 19 of 4,937 genes reach it, and none survives the other filters. 0.995 works and gives 164 genes, but the app’s haplotype-frequency slider steps in units of 0.01, so 0.995 cannot be selected and the next value above 0.99 is 1.00. The arm tightens the gene rule instead — `haplotype_gene_pc = 99`, 176 genes, on-grid, and the same control the preprint already moves to get its published 1,517 figure.

02 R2.3 — Threshold sensitivity

One *filter* is varied at a time; every other filter stays at the published configuration. That is not always one *setting*: the three identity-only rows change the human filter’s combination mode as well as its threshold (`human_id_joint_rule` → “deactivate length filter”, plus `identity_percent` for the $\leq 90\%$ and $\leq 70\%$ rows; the $\leq 80\%$ row changes the mode alone). It is unavoidable, since identity is inert while the length criterion is active, and it is the point of those rows. Peptide counts are peptides belonging to the retained genes, on the paper’s definition. Jaccard is on gene sets — overlap divided by union, so 1.0 means identical catalogues and 0 no shared gene. It falls when two catalogues differ greatly in *size* even where one wholly contains the other, so for the nested comparisons here the “% retained” columns are more informative.

SCENARIO	FILTER	EXACT SETTING	GENES	PEPTIDES	JACCARD VS PUBLISHED	GAINED	LOST
APP	—	application defaults	1,142	41,332	0.222	889	0
M253	—	published baseline	253	10,088	1.000	0	0
HUMAN-RELAXED	human	alignment length ≤ 18 aa	376	14,735	0.673	123	0
HUMAN-STRINGENT	human	alignment length ≤ 12 aa	183	7,261	0.723	0	70
HUMAN-IDONLY-RELAXED	human	identity $\leq 90\%$, length filter off	587	21,554	0.431	334	0
HUMAN-IDONLY-BASELINE	human	identity $\leq 80\%$, length filter off	558	20,706	0.453	305	0
HUMAN-IDONLY-STRINGENT	human	identity $\leq 70\%$, length filter off	506	19,125	0.500	253	0
CONSERVATION-RELAXED	conservation	haplotype frequency ≥ 0.95	417	23,144	0.607	164	0
CONSERVATION-STRINGENT	conservation	99% of peptides at ≥ 0.99 (gene rule)	176	5,007	0.696	0	77
INDEL-RELAXED-FREQ	indel	indel frequency ≤ 0.10	260	10,708	0.973	7	0

SCENARIO	FILTER	EXACT SETTING	GENES	PEPTIDES	JACCARD VS PUBLISHED	GAINED	LOST
INDEL-RELAXED-RULE	indel	95% of peptides must pass	291	13,772	0.869	38	0
EXPRESSION-RELAXED	expression	≥ 1 CPM in ≥ 2 replicates (D4)	596	23,321	0.424	343	0
EXPRESSION-STRINGENT	expression	≥ 5 CPM in all replicates (D4) †	251	10,031	0.992	0	2
ORTHOLOGY-RELAXED	orthology	no orthology requirement	290	11,085	0.872	37	0
ORTHOLOGY-STRINGENT	orthology	<i>P. vivax</i> AND <i>P. berghei</i>	251	10,024	0.992	0	2

† The 5-CPM value is a **prespecified sensitivity-analysis setting, not a biologically validated boundary** between expressed and non-expressed genes, and should be described that way wherever it appears (plan Step 6). CPM here is parasite-relative within libraries dominated by host reads, so its stringency is not constant across stages — see S6 in §05.

Consistency holds throughout, checked by set containment: M253 $\subseteq$ every relaxed catalogue, and every tightened catalogue $\subseteq$ M253, at gene and peptide level across all 15 runs. One label needs care — HUMAN-IDONLY-STRINGENT is named for its identity threshold ($\leq 70\%$, the most stringent of that arm) but is a *relaxation* overall, because switching the length filter off removes a constraint; it gains 253 genes, loses none, and is a superset. No benchmark antigen is retained under any scenario, so that column is omitted here — see §03. Full overlap statistics against both reference catalogues are in `supplementary_table_A_sensitivity.csv`.

What this says

The catalogue is **most sensitive to expression and conservation**: relaxing the Day-4 replicate rule more than doubles it (596 genes, Jaccard 0.42), and dropping conservation to 0.95 adds 164 genes. For **indels and orthology** it depends which axis moves. The indel *frequency* threshold barely matters (+7 genes, +2.8%, Jaccard 0.97) and tightening orthology costs 2 genes, but relaxing the indel *gene rule* to 95% adds 38 (+15.0%, Jaccard 0.87) and dropping orthology altogether adds 37 (+14.6%, Jaccard 0.87). Calling those two “a few percent” would understate them. Composition is stable in the sense that matters: no reasonable setting of any single filter removes a substantial part of the published list except by deliberate tightening.

Scenarios from the draft plan that had to be replaced

AS PLANNED	GENES	Δ	WHY IT WAS REPLACED
Human relaxed — identity $\leq 90\%$	253	0	No effect. Under the joint rule the ≤ 15 aa length criterion is binding at every identity setting from 60% upward, so both planned values land on the baseline. Identity only becomes limiting below 60% (252 genes at $\leq 59\%$, 232 at $\leq 50\%$).
Human stringent — identity $\leq 70\%$	253	0	No effect, same reason.

AS PLANNED	GENES	Δ	WHY IT WAS REPLACED
Indel stringent — ≤ 0.01	253	0	No effect. The gene-level indel metric is a step function: identical for any cutoff from 0.00 to 0.05.
Conservation stringent — =1.00	0	-253	Degenerate. Only 19 of 4,937 genes reach it; none survive the other filters.

WORTH ONE SENTENCE IN THE RESULTS

That the human identity threshold is inert over its upper range while the alignment-length criterion is active is a genuine property of the filter. It is inert from $\leq 60\%$ to $\leq 100\%$, the whole span the plan proposed to vary; below 60% it does bite (252 genes at $\leq 59\%$, 232 at $\leq 50\%$). Holding identity at $\leq 80\%$ and varying length instead gives 183 / 253 / 376 genes at ≤ 12 / ≤ 15 / ≤ 18 aa. Switching the length filter off — an option the app already offers — makes identity meaningful again: 506 / 558 / 587 genes at $\leq 70\%$ / $\leq 80\%$ / $\leq 90\%$.

For indels there is no stringent arm, checked on every axis the filter has. The frequency threshold is inert all the way down: 0.05, 0.04, 0.03, 0.02, 0.01 and 0.00 all return the same 253 genes, because every gene at or below 0.05 sits at exactly 0.0. The gene rule is already at its 100% maximum. And restricting *which* indels are counted **relaxes** the filter — “Only Insertions” 280 genes, “Only Deletions” 260, “Only Frameshifts” 294, insertions-and-frameshifts 297 — since counting fewer indels lowers each gene’s maximum. The filter is **maximally stringent at its default settings**, with no stringent direction available anywhere.

ONE CAVEAT ON THE CONSERVATION-STRINGENT ARM

The stringent conservation arm tightens the **gene rule**, not the threshold. The app's haplotype-frequency control is a slider with min 0.0, max 1.0, step 0.01, so the only selectable values are 0.00, 0.01 ... 1.00: **0.995 is not one of them**, and neither is 0.999. There is no on-grid threshold alternative either, because the next selectable step above 0.99 is 1.00, which empties the catalogue. Earlier drafts used `strain_conservation = 0.995` (164 genes), a real run that could not have been reproduced by clicking the published app; it has been replaced. Holding the threshold at the published 0.99 and moving the gene rule instead keeps every scenario reproducible in the app — here is that axis in the full M253 configuration:

SETTING	COLUMN	GENES	PEPTIDES	ON THE APP'S GRID?
haplotype_gene_pc = 95	_p05	253	10,088	BASELINE
haplotype_gene_pc = 96	_p04	228	8,700	YES
haplotype_gene_pc = 97	_p03	206	6,994	YES
haplotype_gene_pc = 98	_p02	186	5,582	YES
haplotype_gene_pc = 99	_p01	176	5,007	ADOPTED
haplotype_gene_pc = 100	_p00	170	4,791	YES
<i>strain_conservation = 0.995 (superseded)</i>	<i>_p05</i>	<i>164</i>	<i>5,171</i>	NO — OFF-GRID

The adopted arm is **haplotype_gene_pc = 99 — 176 genes / 5,007 peptides**, a strict subset of the published catalogue (0 gained, 77 lost, Jaccard 0.696). What decides it is that the preprint *already* reports a 99% conservation gene rule — it is how the published 1,517-gene figure is obtained — so the Methods can cite a value the analysis already uses, and anyone clicking the published app can reproduce the scenario. Table A, the retention matrix, the 15 frozen configs and `run_outputs/` have been regenerated; the 0.995 run is kept in `rejected_scenarios.csv` with its reason.

03 R2.1 — The vaccine-antigen benchmark

All eight antigens are excluded, under every one of the fifteen configurations. The retained/excluded matrix the plan proposed is therefore a single uniform block, and the "antigens retained" column reads none retained in

every row. Both are uninformative as designed; the reason they come out that way is the finding.

The Discussion already makes this argument in words:

"Previous vaccine efforts have repeatedly encountered issues with limited efficacy due to antigenic variation among parasites, including trials targeting antigens such as circumsporozoite protein (CSP), apical membrane antigen 1 (AMA1), or merozoite surface protein 1 (MSP1). These vaccines have often resulted in strain-specific immune responses..."

The benchmark supplies the numbers behind that sentence. All eight fail on **strain conservation**, and AMA1 and MSP1 — two of the three the paper names — fail by a wide margin on both measures (27.4% and 54.7% of peptides against a 95% requirement; 0.20 and 0.49 against 0.99). CSP needs stating carefully: it misses badly on the threshold it tolerates (0.34 against 0.99) but is among the closest of the eight on the peptide-fraction axis (**74.4%** of its peptides clear 0.99, second only to RH5's 80.8%; the 70 in the table is the gene-rule slider setting, not a peptide fraction). The two axes rank the eight differently, so neither should be quoted alone.

So the answer to "does your tool recover the known antigens?" is no — and it rejects them for exactly the reason the paper already identifies as having limited those vaccines in the field. That second half is an *interpretation* consistent with the computation, though the computation does not establish it. What the computation establishes is narrower: all eight fail the configured conservation criterion, and conservation is the filter responsible in every case. The step to "working as intended" rests on the paper's own argument about antigenic variation. Framed that way it is still the stronger answer to R2.1, and it claims no more than was shown.

Step 18 asks for two things to be stated explicitly, and both hold: an established antigen *"may fail because it is highly polymorphic, despite having demonstrated biological or immunological relevance"* — which is what the percentile ranks in S1-S2 show for all eight — and, conversely, recovery would have *"provide[d] a useful face-validity benchmark for the workflow but does not demonstrate that its retained peptides are immunogenic or protective"*. Nothing here speaks to immunogenicity or protection in either direction.

The reviewer asked about *"licensed and leading"* antigens, and the two are not the same thing. Of the eight, only **CSP** underlies a licensed product — both WHO-

recommended and prequalified malaria vaccines, RTS,S/AS01 and R21/Matrix-M, are CSP-based. The other seven are clinical or preclinical candidates. Describe the panel as the eight the reviewer requested, *one licensed and seven leading candidates*, and read nothing here as evaluating a licensed vaccine.

One caveat on the per-filter failures below, established in S1 of §05:

conservation is the only filter all eight fail, though human similarity independently excludes CSP and indels independently exclude four more. Six of the eight fail the day-4 expression filter, but seven are expressed in sporozoites and substituting the sporozoite criterion retains none of them. Four have no recorded *P. vivax* orthologue — RH5, CSP, EBA175 and LSA3. For RH5 that is expected biology for a *P. falciparum*-restricted gene; for CSP it is not, since a *P. vivax* CSP is well described and is the basis of that species' leading vaccine candidates. That cell reports what the shipped hom_homolog_in_pv column records, not whether a counterpart exists, and the column should be read that way. Count the conservation column, and treat expression and orthology as largely stage-explained.

ANTIGEN	GENE ID	HUMAN NEED 75%	CONSERVATION NEED 95%	MAX FREQ NEED 0.99	INDEL NEED 100%	D4 REPS NEED 3	P. VIVAX	EXCLUDED BY
CSP	PF3D7_0304600	40 ✗	70 ✗	0.34 ✗	100 ✓	3 ✓	no ✗	human, conservation, orthology
AMA1	PF3D7_1133400	88 ✓	25 ✗	0.20 ✗	100 ✓	3 ✓	yes ✓	conservation
RH5	PF3D7_0424100	85 ✓	80 ✗	0.93 ✗	100 ✓	2 ✗	no ✗	conservation, expression, orthology
TRAP	PF3D7_1335900	75 ✓	22 ✗	0.26 ✗	93 ✗	2 ✗	yes ✓	conservation, indel, expression
CeITOS	PF3D7_1216600	88 ✓	40 ✗	0.22 ✗	100 ✓	2 ✗	yes ✓	conservation, expression
EBA175	PF3D7_0731500	85 ✓	70 ✗	0.58 ✗	98 ✗	0 ✗	no ✗	conservation, indel, expression, orthology
MSP1	PF3D7_0930300	85 ✓	50 ✗	0.49 ✗	92 ✗	2 ✗	yes ✓	conservation, indel, expression
LSA3	PF3D7_0220000	85 ✓	60 ✗	0.89 ✗	95 ✗	1 ✗	no ✗	conservation, indel, expression, orthology

ANTIGEN	% OF PEPTIDES PASSING CONSERVATION (REQUIREMENT 95%)	GENE-RULE SLIDER SETTING TOLERATED	CONSERVATION PERCENTILE AMONG ALL 4,937 GENES	IDENTITY PERCENTILE	INDEL PERCENTILE
AMA1	27.4	25	0.0	45.0	30.0
TRAP	22.8	22	0.1	16.7	16.6
CeITOS	50.0	40	0.0	51.5	30.0
CSP	74.4	70	0.3	6.9	30.0
MSP1	54.7	50	0.8	56.7	1.9
EBA175	72.0	70	2.0	50.2	19.2
LSA3	69.7	60	12.6	56.7	7.4
RH5	80.8	80	17.5	56.4	30.0

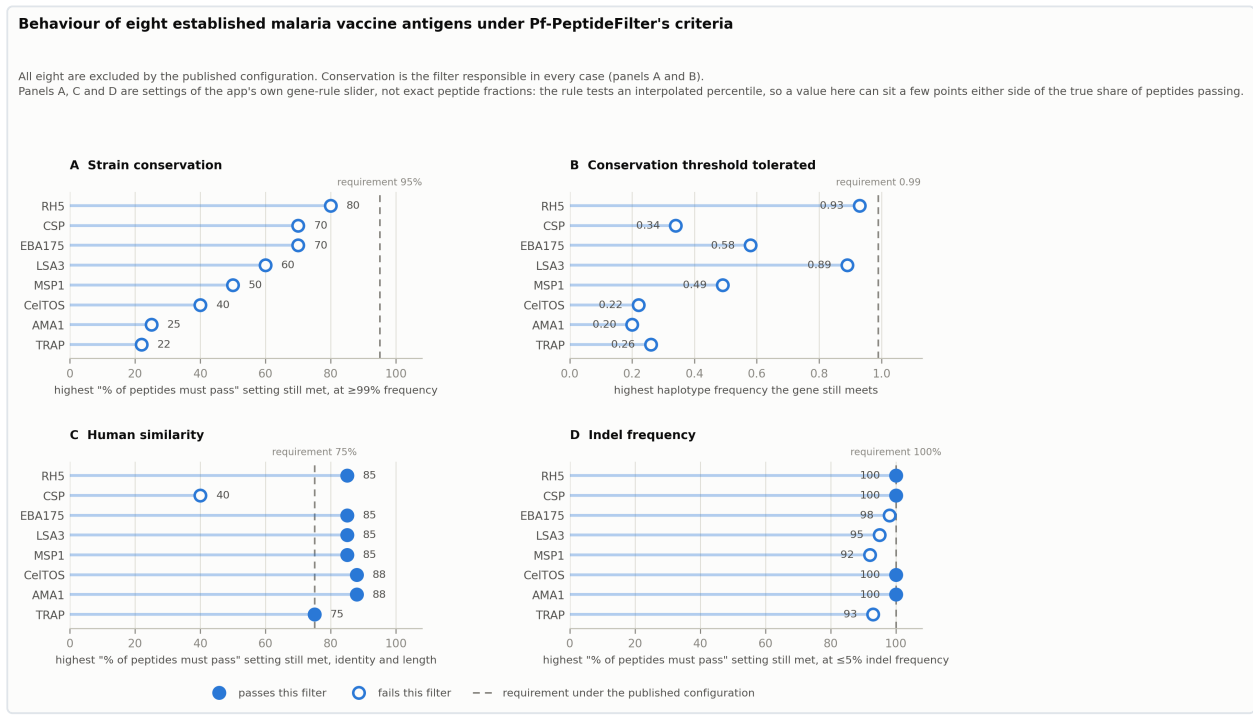
The two quantities the main table cannot carry, both in `supplementary_table_B_antigens.csv`.

Column 2 is what plan Step 14 asked for — the actual proportion of each gene’s peptides satisfying the peptide-level condition, which distinguishes “94% of peptides pass a 95% requirement” from “only 20% pass”. It is *not* column 3, the slider setting; the two differ in 4 of the 24 antigen × filter cells. **Columns 4–6** rank each gene against every other gene carrying that metric — 4,930 for conservation, 4,933 for identity, all 4,937 for indels — read as “better than X% of the genome”, so a low number is a bad gene. They are what makes the benchmark more than a restatement of the filter’s definition: five of the eight are in the bottom 1% of the genome for conservation while sitting mid-range on identity. Sorted by conservation percentile, worst first.

✓ passes that filter, ✗ fails it. Every verdict is a run of the app with all filters neutralised except the one under test. Each percentage is read off the app’s own gene-filtering slider — the highest “% of peptides that must pass” setting at which the gene survives — so the values step through the 39 settings the slider offers rather than being computed independently. Each was checked to be the correct floor on that grid, and the pass/fail sequence is monotone in the setting for all eight genes on all three filters. Read them as slider settings, *not* as the exact share of peptides passing.

Because the gene rule tests a linearly interpolated percentile, the setting can sit either side of the grid floor of the true peptide fraction, and does for **4 of these 24 cells**: CeITOS conservation reads 40 where 50.00% of its peptides are at ≥ 0.99 (grid floor 50); EBA175 human reads 85 where 83.33% pass both conditions (floor 82); TRAP indel reads 93 where 92.98% pass (floor 92); MSP1 indel reads 92 where 91.86% pass (floor 91). No verdict changes, but the numbers should not be quoted as percentages of peptides.

“Max freq” is the most stringent haplotype-frequency threshold the gene still meets at the 95%-of-peptides rule, found by sweeping the slider at its own 0.01 step. The CSV columns holding these values are `*_gene_rule_pct_tolerated`; they were previously `*_pct_peptides_passing`, which asserted exactly the wrong thing. Full definitions for both tables are in `analysis/output/COLUMN-DEFINITIONS.md`.



Proposed supplementary figure. Distance to each threshold, replacing the retained/excluded matrix. It separates AMA1 — which fails conservation by a wide margin on both measures, 25% of peptides and a tolerated threshold of 0.20 — from RH5 and LSA3, which come far closer on the threshold axis (0.93 and 0.89 against a 0.99 requirement). Note that the two panels rank the eight differently: LSA3 is nearly at the requirement in panel B but further from it than CSP in panel A. A uniform grid of "excluded" cannot show that. Colour carries no meaning, so the figure survives greyscale printing and colour blindness; a hollow marker fails the filter, a filled one passes.

04 Changes needed to the manuscript

Why five sentences in print have to change. The gene-level controls are labelled "% of peptides that must pass", but the code selects a precomputed percentile column and tests that — it never counts peptides. Take PF3D7_0308800, one of the 253: ten peptides at haplotype frequencies 0.9876, 0.9960, 0.9960, 0.9987, 0.9990, 0.9990, 0.9991, 0.9995, 0.9996, 0.9999. At a 0.99 threshold nine of ten pass — 90%, short of the 95% rule — so counting would drop the gene. The app takes the 5th percentile instead, interpolates between the lowest two values to 0.9914, and keeps it.

Label and arithmetic agree for most genes and diverge for a minority: **19 of the 253** are retained on fewer than 95% of their peptides actually meeting the threshold. That is what makes the five in-print sentences wrong, why Table B reports the slider setting and the true peptide fraction as separate columns, and why the two must not be converted into one another. The indel filter is the

exception — its rule sits at 100%, so the percentile is the maximum and “every peptide must pass” is literally true.

Six changes, in the order to work through them. Five are in the published preprint; the sixth is in the paragraph drafted for the revision. All six are copy-ready in analysis/MANUSCRIPT-CORRECTIONS.md — sentence as published, replacement beneath. The detail and the numbers behind each follow below.

#	WHERE	THE CHANGE	IN PRINT?
1	Results	The 1,274-gene sentence . The human gene rule is 75%, not 95%, and the sentence attaches 95% to both filters. Run as written it gives 158 genes.	YES
2	Results	The 2,107-gene sentence . The parenthetical says a haplotype present in 95% of isolates; the threshold is 99%.	YES
3	Results	indel frequency < 5% → ≤ 5%. No numerical effect.	YES
4	Results	Homology names <i>P. vivax</i> and <i>P. berghei</i> ; the published run used <i>P. vivax</i> alone. Both gives 251 genes.	YES
5	Methods	“assessed per peptide and averaged across genes ”. It is not a mean but a 75th-percentile test.	YES
6	Methods	The expanded configuration paragraph drafted for the revision. Three clauses do not describe what the code does — 6a, 6b, 6c below, with the full redraft after them.	NOT YET

Changes 1-4 — errors already in print

THE 1,274 SENTENCE — WRONG BY A FACTOR OF EIGHT

"Applying both the strain conservation and human identity filters simultaneously-i.e. by requiring 95% of peptides per gene to exceed a 99% conservation threshold and remain below 80% identity to any human exon with a maximum alignment length of 15 amino acids-reduced the number of retained genes to 1,274"

The human gene rule is **75%**, not 95%; the sentence attaches 95% to both filters. Run as described it gives **158 genes**, not 1,274. And "simultaneously ... and" invites the joint-subset reading, which gives **92**: of the app's 1,274, only 92 have ≥95% of peptides passing all three criteria together — conservation ≥0.99, identity ≤80% and alignment length ≤15 aa. (Drop the length criterion from that reading and it is 1,133, which is why the subset has to be specified.) The 1,274 comes from conservation's 5th-percentile test at ≥0.99 and identity's 75th-percentile test at ≤80% and length's 75th-percentile test at ≤15 aa, each evaluated separately — 1,274 / 158 / 92 for the three readings.

THE 2,107 SENTENCE — THE PARENTHETICAL NAMES THE WRONG THRESHOLD

"... retain only those genes for which at least 95% of peptides exhibited a dominant haplotype (i.e. a haplotype present in **95% of isolates or more**), a total of 2,107 genes and 84,556 peptides were retained"

It should read **99%**. The parenthetical conflates the gene rule (95% of peptides) with the frequency threshold (0.99). At 0.95 the same gene rule returns **3,824 genes**; 2,107 requires 0.99. The counts are right — 2,107 / **84,556** and 1,517 / **42,208** reproduce exactly, and both peptide figures are now asserted by `validate()`. The independently filtered peptide set is **326,701 for both** rows, since peptide-level filtering cannot see the gene rule — further confirmation that every peptide number in the paper is the “belonging to” definition.

TWO SMALLER MISMATCHES IN THE SAME PASSAGE

"indel frequency < 5%" — the code tests ≤ 0.05 . Here it changes nothing: every gene at or below 0.05 sits at exactly 0.0 and the next distinct value is 0.057, so strict and non-strict both give 3,457 genes. Worth aligning anyway, since a reviewer reproducing the work will notice.

"cross-species homology (presence of one-to-one orthologs in *P. vivax*, and *P. berghei*)" — the published configuration requires *P. vivax* only. Requiring both gives **251 genes**, not 253. The very next sentence says "homology with *P. vivax*", so the two disagree with each other.

5 — “averaged across genes” in the published Methods

One further wording point in the published Methods, and it is the same class of error as §04. The Methods say human identity is "*assessed per peptide and averaged across genes*", and Table 1 says "*filtering is performed at the peptide level and summarised per gene*". Neither is a mean: the default rule is a 75th-percentile test. "Averaged" is wrong and "summarised" is too vague to reproduce from — both should name the percentile.

The preprint's *default-thresholds* sentence reproduces the result correctly. It describes the catalogue as "*default thresholds (human identity $\leq 80\%$, strain conservation $\geq 99\%$, indel frequency < 5, homology with *P. vivax*, gene expression in day 4 of liver stage infection)*" — and it reproduces 253 genes and

10,088 peptides exactly. Two details in that sentence and its neighbour still need correcting, though neither changes a number: the operator is $\leq$ rather than $<$, and the preceding sentence names both *P. vivax* and *P. berghei* where the run used *P. vivax* alone. Both are in the “two smaller mismatches” callout below.

Every *count* in the preprint reproduces; it is the prose describing how the counts were reached that is wrong. The two sentences before it, giving the 2,107 and 1,274 figures, do not match the configuration that produces them — and unlike the methods paragraph, they are already in print. The expanded paragraph drafted for the revision has three clauses that do not describe what the code does: one wrong number, and two that describe the gene-level rule in a way the implementation does not support.

6a — the indel gene rule was 100%, not 95%

"For indel filtering, **at least 95% of peptides** within each gene were required to pass the $\leq 5\%$ frequency threshold, considering indels of any type or predicted effect."

WHAT IT ACTUALLY WAS

The indel gene rule was left at the application default of **100% of peptides**. At 95% the catalogue is **291 genes / 13,772 peptides**, not 253 / 10,088. The other two figures in that paragraph — 75% for human identity, 95% for conservation — are correct.

6b — "applied jointly" describes something the app does not do

"For human identity filtering,
at least 75% of peptides within each gene were required to pass, with the identity and alignment-length criteria applied jointly and a maximum alignment length of 15 amino acids."

WHAT IT ACTUALLY WAS

Two *separate* percentile columns are tested and combined with AND: $\text{blt_pident_p75} \leq 80$ and $\text{blt_length_p75} \leq 15$. Each is the 75th percentile of that one metric across the gene's peptides, computed independently of the other. The app never asks whether the *same* 75% of peptides satisfies both conditions.

The distinction changes the catalogue. **10 of the 253 retained genes** have fewer than 75% of their peptides satisfying both conditions at once, so a filter written the way the paragraph describes would drop them — PF3D7_1203200 (70.0% of peptides pass both), PF3D7_0721100 (70.8%), PF3D7_0628500 (71.0%), PF3D7_0722700 (71.4%), PF3D7_1237700 (71.4%), PF3D7_1404400 (72.2%), PF3D7_0512000 (72.7%), PF3D7_1308100 (73.3%), PF3D7_1015100 (74.4%), PF3D7_1343400 (74.5%). Across all 4,937 genes the identity-passing and length-passing peptide subsets differ for **4,846** of them. (4,840 is the count whose passing *fractions* differ; a further 6 genes pass equal numbers of peptides on each criterion but not the same ones — exactly the case the word “jointly” would have to cover.)

Separately: the `human_id_joint_rule` setting itself has **no effect on any result**. Applying two column tests in one nested group or in two sequential steps is the same set intersection, and a run of both options returns byte-identical gene sets (253 under M253, 1,142 under application defaults). The radio button changes only how the app groups the two tests in its own step summary — one row rather than two.

6c — the conservation clause has the same problem

"For strain conservation, **at least 95% of peptides within each gene were required to meet the $\geq 99\%$ frequency threshold**, using the dominant amino-acid haplotype irrespective of whether it matched the 3D7 reference."

WHAT IT ACTUALLY WAS

The code tests `hap_top_hap_freq_p05 ≥ 0.99`. The suffix is `_p{100 - haplotype_gene_pc}`, an inversion the human and indel filters do not have, and the direction is right — the 5th percentile clearing 0.99 is the percentile form of “95% of peptides must pass”. But it interpolates linearly between the two peptides that straddle it rather than counting. **19 of the 253** are kept on fewer than 95% of their peptides at ≥ 0.99 ; PF3D7_0308800 is retained on 90.0% (10 peptides, 9 passing, 5th percentile 0.9914). Genome-wide the effect runs **99 genes admitted** on fewer than 95%, against **4 rejected** despite meeting the count. Two of those four are the interpolation disagreeing with the count — PF3D7_0714500 and PF3D7_1443800, both at exactly 0.95, interpolating to 0.98988 and 0.98982. The other two are dropped because the percentile column is missing altogether (PF3D7_0812200 at 0.95 and PF3D7_1440300 at 1.00, both NaN in every `hap_top_hap_freq_p*` column, and `NaN ≥ 0.99` is False); see the second defect below. Under a strict “more than 95%” reading the split is 117 and 1, so the non-strict reading is the one quoted here.

The indel clause needs no such repair. Its rule is `_p100`, the maximum, where the percentile and the count agree exactly — “all peptides within each gene” is literally true, on all 4,937 genes.

Verified against the four published figures, which still reproduce exactly: only `_p05` gives 2,107 genes (`_p04` gives 1,913, `_p06` gives 2,291), so the inversion is both implemented and used correctly.

One more point: **10,088** is the number of peptides *belonging to* the 253 retained genes, as reported on the app's gene-results page. Pf-PeptideFilter also filters peptides independently, and that count for the same configuration is **84,859**. Both are legitimate; the paper should say which one it means.

Redrafted paragraph

How to read it: GREEN marks the clauses that differ from the paragraph as currently drafted — those are the edits. Grey italic is a placeholder still to be filled. Everything unmarked is the existing text, unaltered.

Drafted by Claude (Opus 5) from the corrections above — a starting point to edit, not finished prose. Highlighted spans are the clauses that changed; the rest is the existing paragraph unaltered.

As an illustrative analysis, we applied the application's default thresholds for human identity ($\leq 80\%$), strain conservation ($\geq 99\%$) and indel frequency ($\leq 5\%$), together with two filters that are not active by default: liver-stage expression and cross-species homology. All remaining settings were left at their defaults.

For human identity filtering, genes were retained where the 75th percentile of peptide identity was $\leq 80\%$ and the 75th percentile of peptide alignment length was ≤ 15 amino acids, each percentile evaluated separately across the gene's peptides; the two criteria therefore apply to the same gene but need not be met by the same subset of its peptides. For strain conservation, genes were retained where the 5th percentile of peptide haplotype frequency was $\geq 99\%$ — the percentile form of the application's "at least 95% of peptides" gene rule — using the dominant amino-acid haplotype irrespective of whether it matched the 3D7 reference. For indel filtering, all peptides within each gene (100%) were required to pass the $\leq 5\%$ frequency threshold, considering indels of any type or predicted effect. For the expression filter, genes were required to have expression of ≥ 1 CPM in all three biological replicates at liver-stage day 4. Day 4 was selected as a representative intermediate liver-stage time point because it provides substantially greater parasite transcript coverage than day 2, while preceding the late liver-stage transcription observed at days 5 and 6 *[ref still needed]*. *[Supported by the application's own expression table, worth citing alongside any external reference — see the note below.]* For the homology filter, we required the presence of a *P. vivax* orthologue, as *P. vivax* is a major human malaria parasite and therefore provides a biologically relevant cross-species comparator, while also illustrating the functionality of the homology filter. Under this configuration, 253 genes were retained, collectively containing 10,088 peptides. Because Pf-PeptideFilter applies gene-level and peptide-level filtering independently, we report here the total number of peptides belonging to the retained genes. This represents a shortlist of potential vaccine targets for downstream

experimental validation rather than a uniquely optimal candidate set.

THE DAY-4 CLAIM IS SUPPORTED BY THE APP'S OWN DATA

The [ref still needed] marker is still a blocker for an external citation, but the claim is no longer resting on one. Counting genes at ≥ 1 CPM in all three replicates in data/gene-metrics-filtering.csv.gz:

LIVER STAGE	GENES ≥ 1 CPM IN ALL 3 REPLICATES	GENES ≥ 1 CPM IN ≥ 1 REPLICATE	MEDIAN CPM (REPLICATES POOLED)
Day 2	36	1,486	0.00
Day 4	1,583	4,364	22.09
Day 5	4,375	4,766	64.77
Day 6	4,657	4,793	67.45

Day 2 is unusable at this rule — 36 genes of 4,937, median CPM 0.00 — while days 5 and 6 are close to saturation (4,375 and 4,657, i.e. 89% and 94% of all genes), at which point the filter stops discriminating. Day 4 at 1,583 genes is the only one of the four that is both well-powered and selective, which is precisely the "representative intermediate" argument, from the data the app ships. The median column pools all replicate values rather than taking the median of each gene's own median, which would read 25.02 at day 4 rather than 22.09; the ordering across stages is the same either way. Cite the table; add the external reference when one is available.

Completeness check on the redraft

All 17 settings in the frozen configuration (analysis/output/configs/config_M253.json) are accounted for: 16 are stated in the paragraph above, and the 17th — human_id_joint_rule — is deliberately not, because it does not change the output. Nothing else is left implicit. Checked setting by setting against the JSON.

SETTING	VALUE	WHERE THE PARAGRAPH STATES IT
human_id_gene_rule	min % of peptides	"the 75th percentile of peptide identity ... and the 75th percentile of peptide alignment length"
human_id_gene_pc	75	"the 75th percentile" — same clause
human_id_joint_rule	Joint	not stated, by design — the setting has no effect on the gene or peptide set (verified: both options return the identical 253 genes). It changes only the app's step summary, so stating it in Methods would imply a constraint that is not applied.
identity_percent	80	"human identity ($\leq 80\%$)"

SETTING	VALUE	WHERE THE PARAGRAPH STATES IT
alignment_length	15	"maximum alignment length of 15 amino acids"
haplotype_gene_rule	min % of peptides	"the 5th percentile of peptide haplotype frequency ... the percentile form of the application's 'at least 95% of peptides' gene rule"
haplotype_gene_pc	95	"at least 95% of peptides" — same clause, via the 100-pc inversion
haplotype_type	Any haplotype	"irrespective of whether it matched the 3D7 reference"
strain_conservation	0.99	"the 5th percentile ... was $\geq 99\%$ "
indel_frequency	0.05	"the $\leq 5\%$ frequency threshold"
indel_gene_pc	100	"all peptides within each gene (100%)" — the corrected clause
indel_type	Any	"indels of any type"
indel_frameshifts	Any	"or predicted effect"
expression_rule	All replicates	"in all three biological replicates"
cpm_cutoff	1	"expression of ≥ 1 CPM"
expression_stage	Liver Stage Day 4	"at liver-stage day 4"
homology_species	<i>P. vivax</i>	"the presence of a <i>P. vivax</i> orthologue"

Proposed Results summary (Step 20)

Step 20 asks for “a short summary” covering five named components. **The passage below was drafted by Claude (Opus 5) from the analysis outputs**, as a starting point to edit rather than finished prose. It is not in the manuscript and no manuscript file has been touched. Every figure in it comes from analysis/output/. **How to read it:** GREEN marks the clauses that are only there because of a finding in §05 — the life-stage caveat, the percentile ranks and the African-only scope. The rest is straightforward reporting of Table A.

To assess how sensitive the illustrative catalogue is to the chosen thresholds, we varied each of the five filters in turn while holding the other four at the published configuration. This gave thirteen single-filter variants, which together with the two reference configurations make fifteen runs in total (Supplementary Table A). Four of the five filters were varied in both directions; the indel filter admits relaxation only, because the published configuration already requires every peptide in a gene to pass and the frequency threshold is inert at every value below 5%. Candidate-set size ranged from 176 to 1,142 genes against a baseline of 253.

Composition was nested in the direction each variant moves: every relaxation retained all 253 published genes, and each of the four configurations that tightened a criterion was a strict subset of them. The catalogue was most sensitive to the expression and conservation criteria — relaxing the day-4 replicate requirement from three replicates to two more than doubled it to 596 genes (gene-set Jaccard 0.42 against the published catalogue), and lowering the haplotype-frequency threshold from 0.99 to 0.95 added 164 genes — and least sensitive to the indel frequency threshold, which added 7 genes (Jaccard 0.97) across its full usable range. The published catalogue is a strict subset of the one produced by the application's default settings: all 253 genes are retained among the 1,142 obtained with the two optional filters inactive, the difference being entirely the addition of the liver-stage expression and *P. vivax* orthology requirements. None of the eight benchmark vaccine antigens was retained under the published configuration, nor under any of the fifteen runs examined (Supplementary Table B). Strain conservation was the criterion responsible in every case: all eight fell below the requirement that a gene's 5th-percentile peptide haplotype frequency reach 0.99 — the percentile form of the application's "95% of peptides" gene rule — with the proportion of qualifying peptides ranging from 22.8% (TRAP) to 80.8% (RH5), and five of the eight ranking within the bottom 1% of all genes on this metric.

Conservation was not, however, the only criterion to exclude these antigens on grounds independent of life stage: applying the human-similarity and indel filters alone returns 2,397 genes and restores only three of the eight, with circumsporozoite protein excluded by human similarity and TRAP, EBA175, MSP1 and LSA3 by indel frequency. Six of the eight did not meet the liver-stage day-4 expression requirement, which for the four blood-stage antigens in the panel reflects the life stage assayed rather than a property of the gene; seven of the eight are expressed in sporozoites, and substituting the sporozoite-stage criterion returned 695 genes without restoring any antigen to the catalogue. These results

indicate that the catalogue's size is materially dependent on the expression and conservation thresholds while its composition is stable under relaxation, and that established vaccine antigens are excluded by this workflow principally because **they are among the most polymorphic genes in the 8,492 African isolates from which the conservation metric is derived** — a consequence of the conservation-first design rather than an assessment of their merit as vaccine targets. **Because conservation is estimated from African genomes only, the catalogue should be understood as conserved with respect to that population rather than globally.**

That covers all five of Step 20's components in order: degree of change in counts and composition; which filters produced the greatest sensitivity; similarity between the M253 and application-default catalogues; which benchmark antigens were retained under M253 (none, stated directly); and which filters excluded the rest. The stage caveat from S1, the percentile ranks from S2 and the African-only scope from S4 are folded in, since each changes what the passage is entitled to claim — it no longer blames the non-conservation failures on life stage alone, four indel failures and one human-similarity failure being stage-independent, and it no longer calls the identity-only rows tightenings, since they gain genes.

Text changes argued elsewhere in this report

Seven further wording points fall out of the analysis rather than from a mismatch with the code, so the argument for each sits in the section that produced it. Listed here so the whole set of manuscript edits can be worked through from one place.

Results

The human identity threshold is inert over its whole upper range while the alignment-length criterion is active — worth one sentence. [§02](#)

Methods

Describe the 5-CPM value as a prespecified sensitivity setting, not a validated boundary (plan Step 6). [§02](#)

Results / Discussion

Describe the panel as the eight antigens the reviewer requested — *one licensed and seven leading candidates* — since only CSP underlies a licensed product. [§03](#)

Results

Quote the genome percentile ranks, not just pass/fail: five of the eight are in the bottom 1% for conservation. [§05](#), S2

Results

Do not present the antigen exclusions as five independent filter failures — three of the five filters discriminate on stage-independent grounds, and conservation is the only one that catches all eight. §05, S1

Methods

Give the day-4 rationale as parasite transcript depth, and note that “≥1 CPM” is not constant stringency across stages. §05, S6

Discussion

State that conservation is measured in African isolates only (8,492 of Pf7’s 16,203 genomes), so the catalogue is African-frequency rather than global. §05, S4

Methods

State the peptide definition once — peptides *belonging to* retained genes — and use it throughout. §05, S8

05 Additional comments

Eight further points that came out of the work, plus one defect found in a shipped data column along the way. Each affects what the write-up can claim.

How the numbers were produced

Every figure comes from a run of the app’s own filter_datasets() in pep/filter.py, against the repository’s own config/filters.json and data/*.csv.gz; no filter condition is reimplemented anywhere. filter_datasets() touches Streamlit in two lines, both plain reads of the settings, so substituting a dictionary lets it run from a script — which is what makes 269 configurations feasible where clicking sliders is not (24 in run_sensitivity.py, 245 in run_antigens.py, counted by instrumenting run()).

Provenance checks against the live repository

Checked against github.com/malariagen/pf-peptidefilter fetched 4 September 2026 — prod at 5861f79, main at 3a9af72 — not against a cached copy. Every app file in the working tree used for the analysis is byte-identical to live main.

COMPONENT	LIVE PROD (5861F79)	WHAT WAS RUN
gene-metrics-filtering.csv.gz	md5 fdee05f9...	IDENTICAL
peptide-metrics-filtering.csv.gz	md5 3380de0f...	IDENTICAL
go-data-core.csv	md5 compared	IDENTICAL
config/filters.json	byte-for-byte	IDENTICAL

COMPONENT	LIVE PROD (5861F79)	WHAT WAS RUN
UI defaults (all thresholds)	UI_CONFIG	IDENTICAL
Filtering logic	peptidefilter/filter.py	differs only in import order and one summary column renamed proportion → fraction; no difference in filtering behaviour
Environment	pandas 1.5.1	PINNED TO MATCH

Reproduction of published figures

RESULT IN THE PREPRINT	PUBLISHED	REPRODUCED	
Conservation alone, 95% of peptides at ≥99%	2,107 genes	2,107	MATCH
Conservation at 99% of peptides	1,517 genes	1,517	MATCH
Conservation + human identity	1,274 genes	1,274	MATCH
All five filters (the illustrative catalogue)	253 / 10,088	253 / 10,088	MATCH

These four are run as an assertion at the start of every analysis script (pepfilter_headless.validate()); the analysis aborts if any of them drifts.

A DEFECT IN A SHIPPED COLUMN — NUM_PEPTIDES IS SHORT BY ONE

The num_peptides column in the app's own gene-metrics-filtering.csv.gz is **exactly one less** than the gene's true peptide count for **4,423 of 4,937 genes**; the difference is −1 or 0 and never anything else. Over the whole table it gives 374,902 against 379,325 actual peptide rows, and over the published catalogue **9,861 against the correct 10,088**.

No published figure is affected — every peptide count comes from counting rows, and validate() confirms 10,088 independently. But the column is copied into all 15 per-run gene catalogues in run_outputs/, so anyone summing it lands 227 peptides away from the paper. Drop or recompute it before analysis/ is deposited anywhere citable. The defect is upstream, and worth reporting to the app's maintainers.

A SECOND DEFECT, THIS ONE IN A COLUMN THE FILTER READS — SEVEN GENES HAVE NO CONSERVATION PERCENTILE

All 40 hap_top_hap_freq_p* columns are NaN for seven genes. For three of them (PF3D7_1038400, PF3D7_1335300, PF3D7_1335400) no peptide carries a haplotype frequency, so there is nothing to summarise. For the other four there is. PF3D7_1440300 has 45 peptides and **every one of them is at ≥ 0.99** (minimum 0.9917), so its 5th percentile is unambiguously above the threshold

— and the gene is still dropped by the conservation filter, because $\text{NaN} \geq 0.99$ is False. PF3D7_0812200 (95.0% of peptides at ≥ 0.99), PF3D7_0812220 (93.1%) and PF3D7_1440200 (92.9%) are the same case.

No published figure moves: none of the seven enters the 253 on other grounds, and none is a benchmark antigen. But unlike num_peptides this is a column the filter actually tests, and it is the mechanism behind two of the four genes in §04's percentile-versus-count comparison. Worth reporting upstream alongside num_peptides.

S1 · LIFE-STAGE MISMATCH IN THE ANTIGEN PANEL

The panel mixes stages: CSP, TRAP, CelTOS and LSA3 are pre-erythrocytic; AMA1, MSP1, RH5 and EBA175 are blood-stage. M253 requires **liver-stage day-4 expression** and six of the eight fail it, which on the face of it asks merozoite antigens to be expressed in liver stage.

Failure does not track stage, though: CSP and AMA1 both pass the day-4 filter, while LSA3, a liver-stage antigen, fails at ≥ 1 CPM in only 1 of 3 replicates (11.04, 0, 0). And the headline does not rest on those filters. The sporozoite-stage filter gives 695 genes and retains none of the eight; removing expression and orthology entirely gives 1,142 and retains none.

All eight fail strain conservation, which is stage-independent.

The per-filter reasons still need the caveat. Seven of the eight pass sporozoite expression in all five replicates (EBA175 at 2 of 5), so those failures are largely a stage artefact, and RH5 is *P. falciparum*-restricted so failing a *P. vivax* requirement is expected. But conservation is not the only stage-independent discriminator: drop it, keep human similarity and indels, and 2,397 genes return with only **three of the eight** re-entering (AMA1, RH5, CelTOS). Human similarity excludes CSP; indels exclude TRAP, EBA175, MSP1 and LSA3 (MSP1's maximum indel frequency is 0.817 against 0.05). Three of the five filters discriminate on stage-independent grounds — conservation is just the one that catches all eight.

S2 · IS THE BENCHMARK CIRCULAR?

The tool selects for conserved targets and these eight are famous for antigenic variation — the paper's own Discussion says so — so a benchmark that rejects them risks confirming only that the filter does what it is defined to do. As a pass/fail table that objection largely stands.

Percentile ranks blunt it, because the result becomes graded rather than binary. It does not dissolve the objection — the metric being ranked is still the one the filter selects on — but the answer could have come out otherwise. On conservation the eight are extreme: AMA1 and CelTOS beat **0.0%** of the genome, TRAP 0.1%, CSP 0.3%, MSP1 0.8%, EBA175 2.0%, LSA3 12.6%, RH5 17.5%. **Five sit in the bottom 1%** — AMA1 0.02%, CelTOS 0.04%, TRAP 0.14%, CSP 0.30%, MSP1 0.83% — with EBA175 sixth at 2.03%. (4,930 genes carry a conservation value, 4,933 an identity value, all 4,937 an indel value.)

The scales are not comparable, and that needs saying. Ties at the best value cap what any gene can score: conservation reaches **99.6%**, human identity only **56.7%** (2,138 genes tie) and indel frequency **30.0%** (3,457 tie). The eight span 6.9–56.7% on identity, six at or near the ceiling, and four sit exactly at the indel ceiling (CSP, AMA1, RH5, CelTOS); the other four are below it — EBA175 at 19.2%, TRAP 16.6%, LSA3 7.4%, MSP1 1.9%. So on identity and indels most are about as good as a gene can be; only on conservation are they extreme. The columns are in Table B (*_genome_percentile).

S3 · ONE FILTER AT A TIME UNDERSTATES SENSITIVITY BY ABOUT 2.5×

The plan scoped this deliberately: *“This is by no means an exhaustive analysis but we are not going to iterate through hundreds of combinations”*. A factorial sweep is not the remedy, but the write-up should not imply the design explores the space.

Interactions are measurable. Identity is inert only because alignment length binds. Tightening conservation to the 99% gene rule costs **77 genes** at the default human setting but **119** when the human gene rule is loosened to 50%. Some pairs do not interact: relaxing orthology and relaxing the indel gene rule add disjoint sets and combine exactly additively ($328 = 253 + 37 + 38$).

In aggregate the compounding is large. Relaxing all five at once gives **2,048 genes** against an additive prediction of 958 from arms gaining 705 between them — 1,795 where one-at-a-time predicts 705. That uses the indel *gene-rule* arm (`alignment_length ≤ 18`, `strain_conservation ≥ 0.95`, `indel_gene_pc = 95`, `≥ 2 D4 replicates`, no orthology); the indel *frequency* arm (`≤ 0.10`) gives **1,565** against 927. Both are cross-checks outside Table A, recorded in `analysis/output/interaction_checks.txt`. Conjunctive filters behave this way — relaxing one admits only genes already passing the other four — so the single-filter table bounds sensitivity from below. One extra all-relaxed row would bound the space for one run. Beyond what the plan asked for.

S4 · CONSERVATION IS MEASURED IN AFRICAN ISOLATES ONLY

Strain conservation comes from haplotype frequencies in **8,492 African isolates**, a subset of Pf7’s 16,203 genomes. The paper states the scope in Methods, in Table 1 and in the Results, and the plan’s Step 4 confirms that the 1.00 setting “requires the selected haplotype to be observed in all callable **African** samples”. What neither the paper nor this report does is flag it as a *limitation*.

It is one. Conservation in one continental population says nothing about conservation elsewhere, so the catalogue is an African-frequency catalogue and should be described as one. That is a limitation of the data, not a claim about any antigen’s geographic allele-frequency structure. It belongs in the Discussion in a sentence.

The sensitivity analysis cannot address it: the app ships a single African-derived conservation metric, so there is no second geography to vary.

S5 · THE 0.99 THRESHOLD CANNOT BE CHECKED AGAINST SAMPLING ERROR

Each peptide's haplotype frequency is an estimate from a finite sample, so a hard cut at 0.99 moves genes for statistical as well as biological reasons. That cannot be assessed from the shipped data: **neither table carries a coverage or callable-sample count**, so there is no denominator. Nor are the frequencies integer multiples of $1/8,492$ — only 14% are consistent with that, and the observed granularity is $\sim 1.4 \times 10^{-8}$ — so no binomial interval can be constructed.

What can be measured is fragility to the threshold. Of the 253 retained genes, **18 have a 5th-percentile haplotype frequency in [0.990, 0.991]** — within 0.001 of the cut — 33 within 0.002 and 89 within 0.005; the closest sits at 0.99007. Genome-wide, 1,218 genes fall in [0.985, 0.995). A sub-1% movement reshuffles a material fraction of the catalogue — a caveat on the precision of “253”.

S6 · “DAY 2 IS UNUSABLE” IS A LIBRARY-DEPTH ARTEFACT

The day-2/4/5/6 comparison is correct as counts, but the inference drawn from it was wrong. There is no library-size column; the totals are diagnostic. Summed CPM per replicate: **day 2 — 679,169 / 433,162 / 15,682**; day 4 — 755,198 / 1,012,068 / 537,141; day 5 — 913,866 / 1,019,839 / 974,892; day 6 — 834,453 / 801,665 / 949,551; sporozoite — 1.8 to 5.9 million. CPM summed over a transcriptome should be 10^6 . It is, at days 5 and 6. It is not at day 2, and one day-2 replicate totals **15,682 — 1.6% of a day-5 replicate**. These are parasite CPM within libraries dominated by host hepatocyte reads, so at day 2 there is almost no parasite material to measure: 3,367 of 4,937 genes are zero in all three day-2 replicates, against 46 at day 5 and 8 at day 6. Since the rule requires all three replicates, the day-2 figure of 36 genes is effectively set by that one near-empty replicate. Day 4 still stands, for a different reason than given: day 2 cannot support a replicate-based filter *in these libraries*. It also means **“ ≥ 1 CPM” is not constant stringency across stages** — totals differ by up to $\sim 380\times$ — so cross-stage readings, and the ≥ 5 CPM arm, are depth-relative. M253 uses day 4 alone, so the published result is internally consistent.

S7 · STATISTICS AND REPORTING

No inferential statistics are used, which is right for a descriptive sensitivity analysis; nothing claimed would need a test, a p-value or a confidence interval.

Every scenario is nested with respect to M253, verified by set inclusion at gene and peptide level for all 15. Containment is more interpretable than Jaccard for nested sets, and Table A reports both. **The trap:** M253's Jaccard against APP is 0.222, which invites the reading that the catalogues disagree. They do not — M253 is a perfect subset and 100% of its genes are retained. The 0.222 is entirely the 889-gene size difference. Read containment whenever sets are nested and differently sized.

The replacement scenarios were chosen after seeing the originals fail, which is legitimate exploratory work. It is disclosed in `rejected_scenarios.csv` and in departure 3 above.

S8 · TWO PEPTIDE DEFINITIONS

Three numbers exist for the same runs, all legitimate: 10,088 peptides *belonging to* the 253 retained genes; 84,859 for the independently filtered peptide set at the same configuration; and 326,701 for *both* conservation-only rows, because peptide-level filtering strips the percentile suffix and cannot see the gene rule. Table A uses the belonging definition throughout and `validate()` gates 84,556 and 42,208 on it, which is what the paper reports. State the definition once in Methods and use it everywhere.

06 Proposed outputs

Supplementary Table A

The 15 configurations above — sizes, overlap with both reference catalogues, antigen retention. `supplementary_table_A_sensitivity.csv`

Supplementary Table B

The eight antigens, their per-filter values and PASS/FAIL calls, and the filters responsible for exclusion. `supplementary_table_B_antigens.csv`

Supplementary Figure

Distance to threshold, as above. `supplementary_figure_antigen_thresholds.pdf`

Dropped

The 8×12 retained/excluded matrix the plan proposed — the artefact actually generated, `antigen_retention_matrix.csv`, is 8×15 , the three extra human arms having been added after the plan was written. Uniformly "excluded", so it carries no information. The CSV is kept for completeness.

Also kept

A short table of the five scenarios tested and rejected or replaced, so the choices remain traceable. `rejected_scenarios.csv`

Per-run archive

The frozen settings for each of the 15 runs (`configs/`, 15 JSON files) and the gene-level and peptide-level catalogue for each (`run_outputs/`, 45 files).

Provenance record

Re-runnable check of code and data against the live repository. `analysis/verify_provenance.sh` → `provenance.txt`

This document

`analysis/report.html`, rendered to PDF by `analysis/make_report_pdf.py`. Keep the two together: the PDF cannot be regenerated without the source. The renderer is pinned in `analysis/requirements-report.txt` (WeasyPrint 70.0) and deliberately kept out of the repository's `requirements.txt`, which is the app's dependency set and is byte-compared against live main. Pagination is version-dependent, so page numbers should not be cited without the pin and a fresh `pdftinfo` check.

WHAT "REPRODUCIBLE" DOES AND DOES NOT COVER

The *app* and *data* the analysis ran against are verified byte-identical to live main (§05). The analysis itself is tracked as well: `analysis/` is on the `r2-sensitivity-analysis` branch — the scripts, the frozen configurations and all 71 output files — so anyone with access to that branch gets the analysis alongside the app and the data. Only `analysis/private/` is excluded, which holds the plan, the reviewer document and working notes. Two things are still open. A branch is not a citable home, so if the revision cites this analysis it needs merging, archiving with a DOI, or attaching as supplementary material. And the report quotes the plan and the covering email verbatim in places, which is worth a read before the branch goes anywhere public.

Two smaller limits. The eight antigen identifiers were checked against the `gene_name` column of the app's own gene table, which is where the analysis resolves them, rather than against a live PlasmDB query, which needs an API key. And the deployed Streamlit instance exposes no commit metadata, so the comparison is against live prod at 5861f79, not whatever the public URL is currently serving.

07 Open questions

1. **The human filter has two defensible sensitivity arms and I have run both** — varying alignment length, and varying identity with the length filter switched off. **Recommendation: keep both.** Five rows rather than two, but together they establish that identity is non-limiting under the default rule, which is worth a sentence in the Results. Drop the identity-only arm if the table needs to be tighter.
2. **The indel filter gets a relaxation-only pair**, with a sentence explaining that no stringent setting exists. **Recommendation: keep it that way.** Presenting a "stringent" setting that silently equals the baseline would be worse than stating the limit.
3. **Conservation stringent — now `haplotype_gene_pc = 99`, ADOPTED and regenerated throughout. **176 genes / 5,007 peptides**, a strict subset of the published catalogue (0 gained, 77 lost, Jaccard 0.696), replacing `strain_conservation = 0.995` (164 genes), which the app's 0.01 slider cannot select. The reasoning and the full gene-rule axis are in §02.**

4. **Six wording corrections are needed, and five are already in print.** All six are set out as a numbered checklist in §04, and copy-ready in `analysis/MANUSCRIPT-CORRECTIONS.md`.

And a documentation bug in the deployed repository, which is worth reporting to whoever owns it. The README **on prod** — the branch the Streamlit app is deployed from — says at line 17: "*Please use requirements.txt and Python 3.12*". But `requirements.txt` pins `pandas==1.5.1`, which publishes no cp312 wheel (PyPI has cp38, cp39, cp310 and cp311 only), so it cannot be installed on the Python the README asks for without building from source. Anyone following the README's own instructions will fail at `pip install`. Either the pin or the instruction needs to move; 3.11 works today. Verified against a fresh clone of prod at 5861f79.

The branch matters: `main`'s `README.md` is a four-line stub that mentions no Python version, and the working tree used for this analysis matches `main`. The claim is about `prod`.

Scripts: `analysis/pepfilter_headless.py` (app driver and validation), `analysis/run_sensitivity.py`, `analysis/run_antigens.py`, `analysis/make_figure.py`, `analysis/run_interaction_checks.py`, `analysis/run_report_figures.py` (every figure quoted in this report that the other scripts do not produce), `analysis/verify_conservation_grid.py`, `analysis/verify_provenance.sh`, and `analysis/make_report_pdf.py`. Outputs in `analysis/output/`. Scenario definitions were frozen before any output was inspected, and four were then *replaced* after their outputs were seen — three because they changed nothing and one because it emptied the catalogue, which is a departure from Step 19 and what the plan asked for at Step 2. All five rejected or superseded options, with reasons, are in `rejected_scenarios.csv`. See departure 3 in §01.